

Overview: Singular Value Decomposition

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Singular value decomposition (SVD) is perhaps the most useful matrix decomposition in linear algebra (at least from a data scientist's point of view). We will walk through how it can be computed and interpreted.

Let us define our data matrix, $\underline{X} \in \mathbb{R}^{n \times m}$, as a collection of column vectors:

$$\underline{X} = \begin{bmatrix} | & | & & | \\ x_1 & x_2 & \dots & x_m \\ | & | & & | \end{bmatrix}$$

Here, each column represents the observations from one "unit" of data. For example, if we have data on different people, x_i would be the data for the i th person — note that this is different than how we normally think of data, where the rows are a "unit" and the columns are different information about that unit.

What the SVD allows us to do is to take $\underline{X}$ and represent it as the product of three other matrices.

$$\underline{X} = \underline{U} \underline{\Sigma} \underline{V}^T = \begin{bmatrix} | & | & \dots & | \\ u_1 & u_2 & \dots & u_n \\ | & | & & | \end{bmatrix} \begin{bmatrix} \sigma_1 & & 0 \\ & \sigma_2 & \\ 0 & & \ddots \\ & & & \sigma_n \\ \hline & & & & 0 \end{bmatrix} \begin{bmatrix} | & | & \dots & | \\ v_1 & v_2 & \dots & v_m \\ | & | & & | \end{bmatrix}^T$$

The $\underline{U}$ & $\underline{V}$ matrices are unitary orthogonal matrices, and $\underline{\Sigma}$ is a diagonal matrix. ~~U is~~ $\underline{U}$ is $n \times n$, $\underline{\Sigma}$ is $n \times m$, and $\underline{V}$ is $m \times m$.

Some mathematical things worth noting:

- The columns of $\underline{U}$ have the same shape as the columns of $\underline{X}$. If each unit in $\underline{X}$ has 1,000 features, then each u_i is $1,000 \times 1$.

- One way to think of this is that your columns of U are your "eigen" units, arranged such that u_1 is more important than u_2 , u_2 than u_3 , etc., in their ability to describe the variance in the columns of X . They give us a basis we can use to represent each column of X .
- U and V are unitary — this means $UU^T = U^T U = I_n$, and $VV^T = V^T V = I_m$. Their columns are all orthogonal & unit length. This means U provides a complete basis for $\mathbb{R}^n$, which is the subspace for where our data lives.
- Σ is diagonal, non-negative, and σ_i 's are ordered in decreasing magnitude: $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_m \geq 0$. This means the first columns of U and V , u_1 & v_1 , which correspond to σ_1 , are more important than the second columns, and so on and so forth, in describing the information in the X matrix. The relative importance of u_i & v_i are given by the size of their corresponding "singular value" σ_i .

For these reasons, we call U the "left singular vectors", V the "right singular vectors", and Σ the matrix of "singular values".

Because $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_m$, we can ignore some of the smaller ones on the end and approximate X only in terms of the first few "dominant" singular values and vectors.

Lastly, for interpretation: we already know that Σ tells us how important each u_i and v_i is relative to the other columns in U & V .

But what do U & V tell us/mean? Recall that $u_i \in \mathbb{R}^n$ and $x_i \in \mathbb{R}^n$, with x_i being a "unit" (such as a person, or a house if X is about predicting housing prices, etc.) and u_i being an "eigen" unit $\rightarrow$ this also representing (you can think of it as) a person (in dataset with different features for each person) or a house (in housing price dataset). The columns of V are a little less intuitive, and can essentially be thought of as the mixture of all of the "eigen" units needed to make them add up to the original units ~~the~~ (ie, how much of each original unit can be described by each basis vector). It is more helpful to think of V in terms of its transpose here:

$$V^T = \begin{bmatrix} v_1^T \\ v_2^T \\ \vdots \end{bmatrix}$$

$\downarrow$
mixture of eigen units
needed to exactly recover
 x_i , etc.

Last mathematical notes/properties:

- Guaranteed to exist
- Always unique (up to ~~the~~ flipping signs of u_i & v_i vectors)